**Week 13**

# MICROECONOMIC THEORY
## ECON 323 502/503

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**Review**

In **an equilibrium in dominant strategies**, each player is playing a dominant strategy, i.e., a strategy that is optimal regardless of what the opponents do.

A **Nash Equilibrium** is a profile of strategies in which each player is playing a best response to the other player's strategy.

An **equilibrium in maximin strategies** is a profile of strategies in which all players play a maximin strategy. A **maximin strategy** of a player is a strategy that maximizes the minimum gain that can be earned.

# 13.5 Sequential-move Games

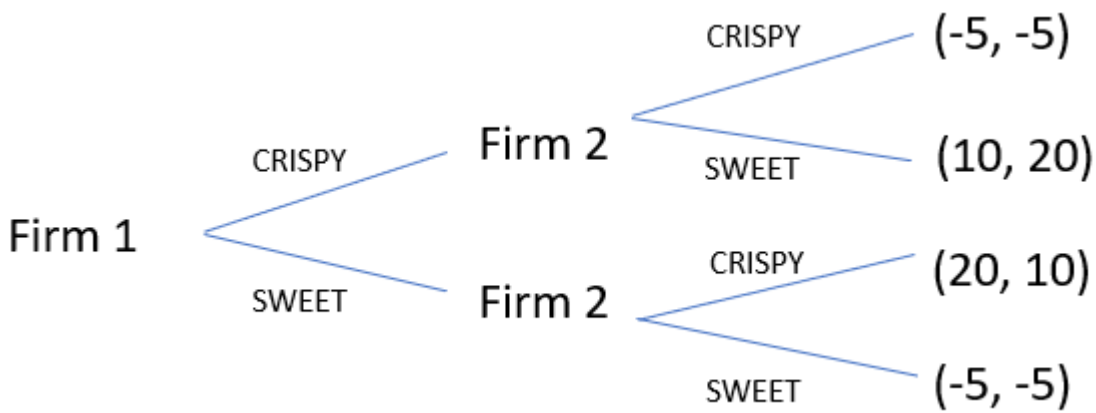
The timing of the game matters!!! Think about the rock-paper-scissor game.

- **Sequential-move game** Game in which players move in turn, responding to each other's actions and reactions.

| | | FIRM 2 | |
|---|---|---|---|
| | | CRISPY | SWEET |
| FIRM 1 | CRISPY | –5, –5 | 10, 20 |
| | SWEET | 20, 10 | –5, –5 |

- If Firm 1 takes its action first. After observing Firm 1's action, Firm 2 moves. What do you think will happen? Suppose players are both rational.

Sequential-move games are often easier to visualize using a game tree, as it clearly represents the timing of actions.




A **strategy** is a contingent <u>plan of actions</u> to take.

Firm 2 can observe Firm 1's strategy and takes an action accordingly, thus each strategy of Firm 2' has the following form --

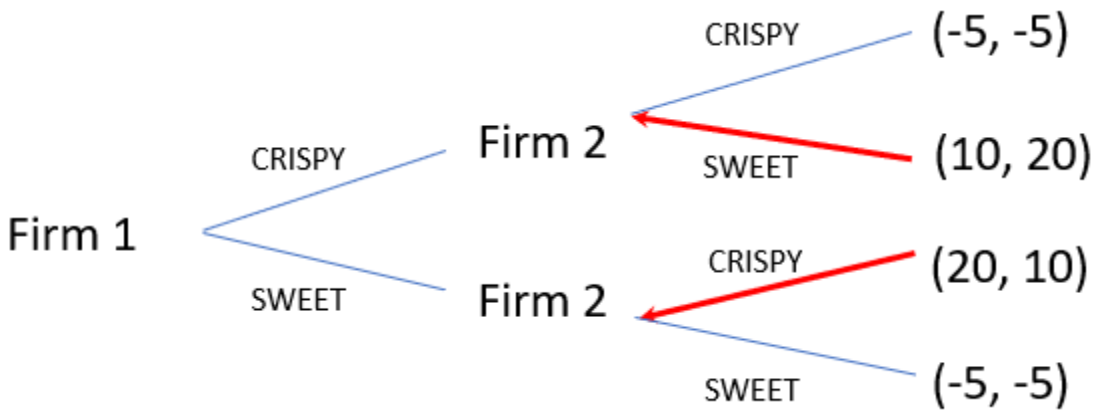
***if Firm 1 plays Crispy, I will play ____; if Firm 1 plays Sweet, I will play ____.***

Firm 2 has **four possible strategies** in total:

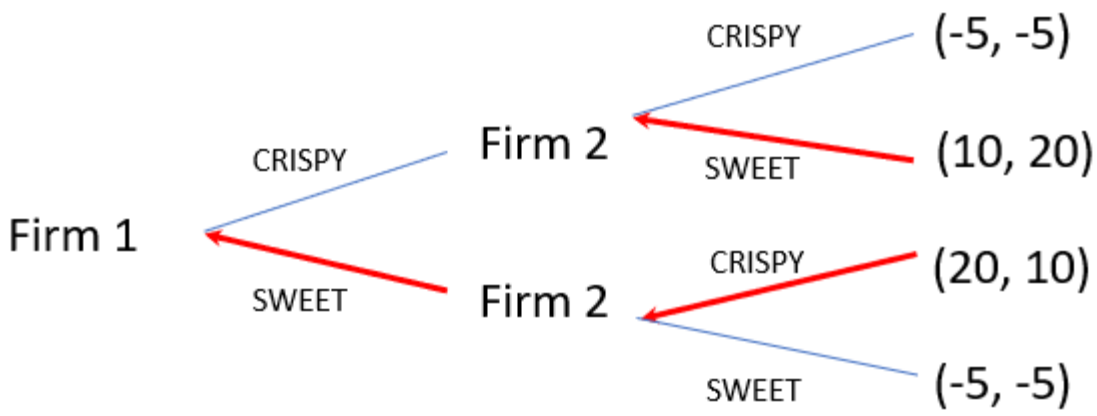
- If Firm 1 plays Crispy, I will play Crispy; if Firm 1 plays Sweet, I will play Crispy.
- If Firm 1 plays Crispy, I will play Crispy; if Firm 1 plays Sweet, I will play Sweet.
- If Firm 1 plays Crispy, I will play Sweet; if Firm 1 plays Sweet, I will play Crispy.
- If Firm 1 plays Crispy, I will play Sweet; if Firm 1 plays Sweet, I will play Sweet.

Firm 1 has **two possible strategies**:

- Crispy
- Sweet




Firm 2's Optimal Strategy: **if Firm 1 plays Crispy, I will play <u>Sweet</u>; if Firm 1 plays Sweet, I will play <u>Crispy</u>**.




Firm 2's Optimal Strategy: **if Firm 1 plays Crispy, I will play <u>Sweet</u>; if Firm 1 plays Sweet, I will play <u>Crispy</u>**.

Anticipating Firm 2's optimal strategy, Firm 1's will choose **<u>Sweet</u>**.

This how we find a **Subgame Perfect Nash Equilibrium** in a sequential-move game via backward induction.

In sequential-move games, we adopt the solution concept of **Subgame Perfect Nash Equilibrium** (in this class), in which:

1. the follower plays its optimal strategy, i.e., takes <u>the optimal action in response to every possible action taken by the leader</u>;

2. <u>anticipating that the follower will play the optimal strategy</u>, the leader chooses its own optimal strategy.

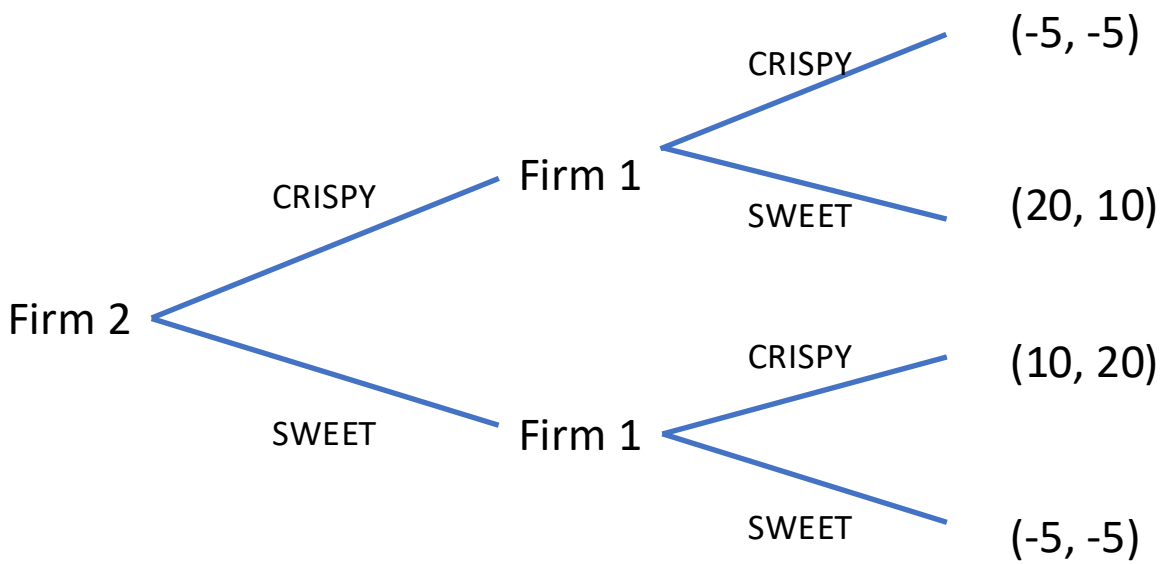
# What if Firm 2 moves first instead?

Will the Subgame Perfect Nash Equilibrium payoffs be different?

Yes! The Subgame Perfect Nash Equilibrium payoffs will be different.

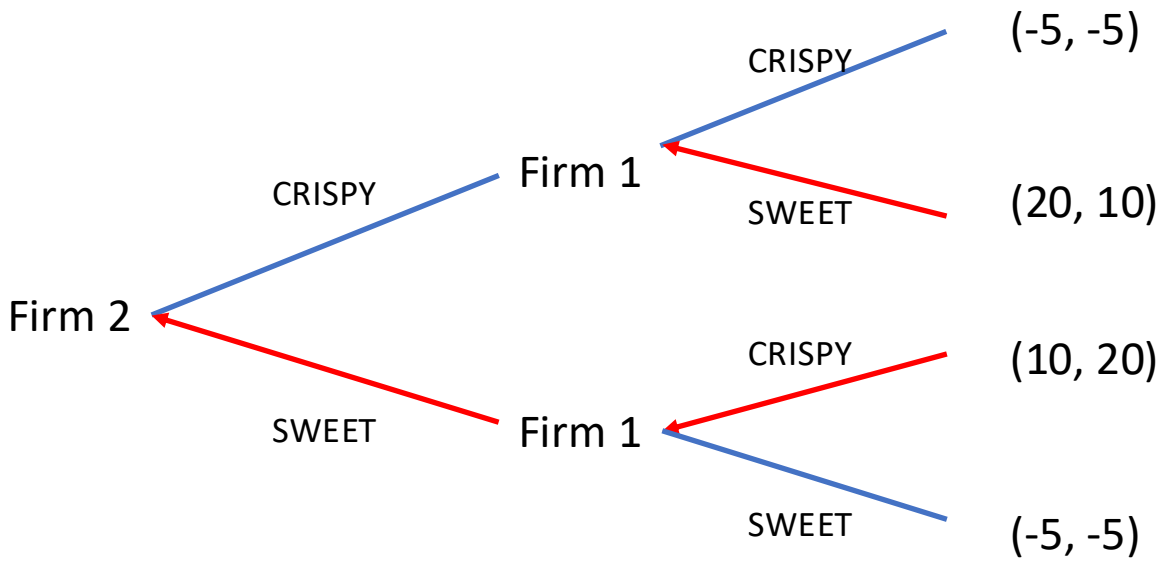
| | | FIRM 2 | |
|---|---|---|---|
| | | CRISPY | SWEET |
| FIRM 1 | CRISPY | –5, –5 | 10, 20 |
| | SWEET | 20, 10 | –5, –5 |

In the following figure, we have **Firm 2 move first**, and we still have the first dimension refer to the payoff of Firm 1 (Row Player).



| | | FIRM 2 | |
|---|---|---|---|
| | | CRISPY | SWEET |
| FIRM 1 | CRISPY | –5, –5 | 10, 20 |
| | SWEET | 20, 10 | –5, –5 |

In the following figure, we have **Firm 2 move first**, and we still have the first dimension refer to the payoff of Firm 1 (Row Player).



## Practice Question 1

In the sequential-move game, if the girl moves first and then the boy follows, what is the equilibrium payoff vector in the subgame perfect Nash equilibrium? (The first element of a payoff vector refers to the payoff of the girl.)

| | | Boy | |
|---|---|---|---|
| | | Movie | Football |
| Girl | Movie | 3, 2 | 0, 0 |
| | Football | 0, 0 | 2, 3 |

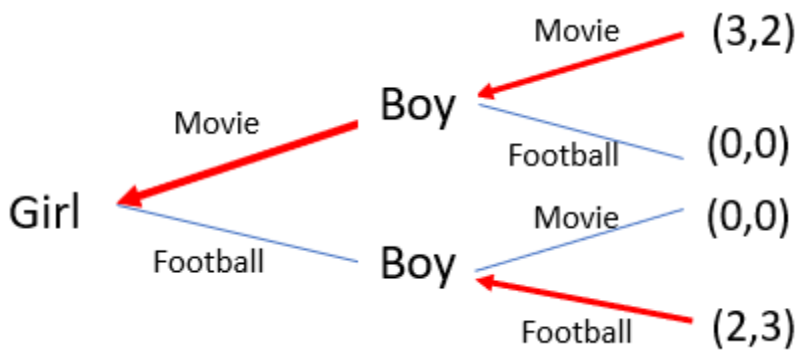
A. (3,2)
B. (2,3)
C. (0,0)

# Practice Question 1 Solution

In the sequential-move game, if the girl moves first and then the boy follows, what is the equilibrium payoff vector in the subgame perfect Nash equilibrium? (The first element of a payoff vector refers to the payoff of the girl.)

| | | Boy | |
|---|---|---|---|
| | | Movie | Football |
| Girl | Movie | 3, 2 | 0, 0 |
| | Football | 0, 0 | 2, 3 |

A. (3,2)
B. (2,3)
C. (0,0)

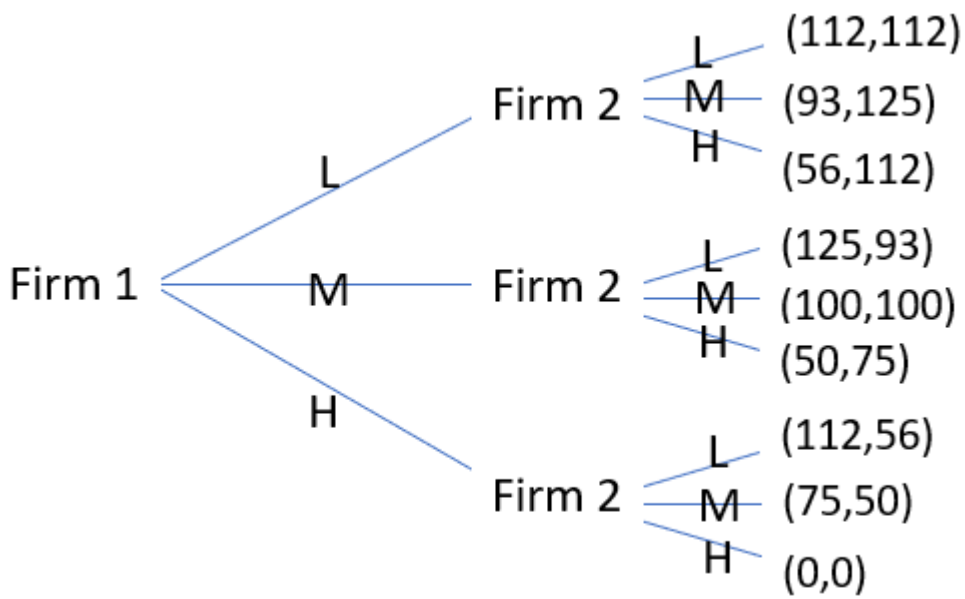



Solution: The boy can observe the girl's action, and thus his optimal strategy is to follow the girl. Anticipating that, the girl chooses to play "Movie". The equilibrium payoff vector is (3,2).

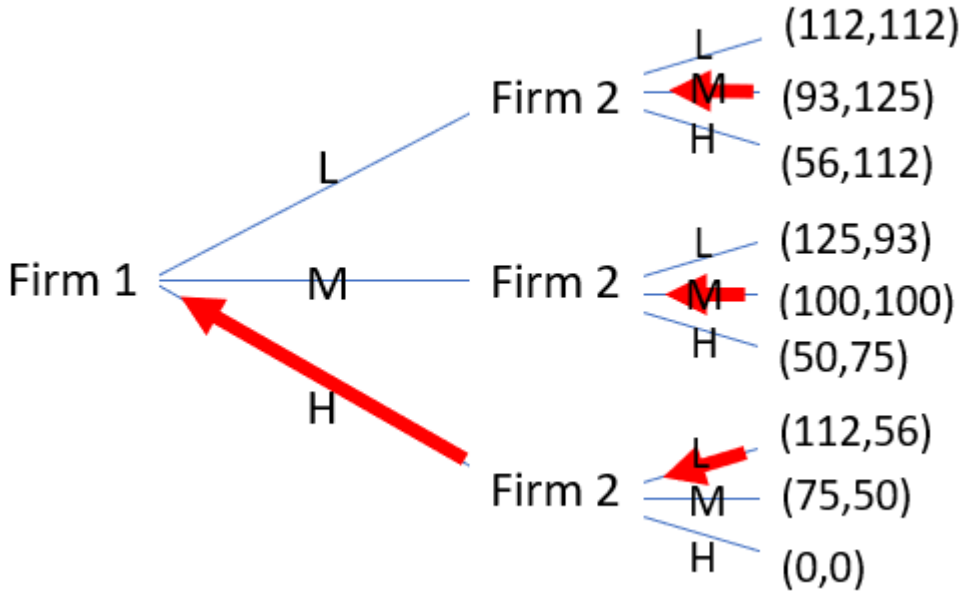
# Duopoly Game: Sequential-Move Version

In a sequential-move game where Firm 1 moves first and Firm 2 follows, what is the subgame perfect Nash equilibrium?

| | | FIRM 2 | | |
|---|---|---|---|---|
| | | **L** | **M** | **H** |
| **FIRM 1** | L | 112, 112 | 93, 125 | 56, 112 |
| | M | 125, 93 | 100, 100 | 50, 75 |
| | H | 112, 56 | 75, 50 | 0, 0 |



# Duopoly Game: Sequential-Move Version Solution




The subgame perfect Nash equilibrium:

- Firm 2: if Firm 1 plays L, I will play M; if Firm 1 plays M, I will play M; if Firm 1 plays H, I will play L.
- Firm 1: H.

The subgame perfect Nash equilibrium payoff profile is (112, 56).

# Duopoly Game: Sequential-Move Version. Variants.

If Firm 2 moves first, and Firm 1 moves after observing Firm 2's action: the subgame perfect Nash equilibrium payoff profile is (56, 112), where the first dimension refers the payoff of Firm 1 (the row player).

In a sequential-move duopoly game (will be called Stackelberg game), there is a first-mover advantage.

In the simultaneous-move variant of the duopoly game (will be called Cournot game), the only Nash equilibrium involves both player playing M.

# First-Mover Advantage

In all three games we study today, the crispy-sweet game, battle of the sexes game, and the sequential-move duopoly game, there is a first-mover advantage.

Is it true that in all sequential-move game, there is a first-mover advantage?

No. Think of the sequential-move rock-paper-scissors game, there is no first-mover advantage.

# A Sequential-Move Game: Divide and Choose

**Players:** Player 1 (Cutter) Player 2 (Chooser)

**Game Structure:**

- Player 1 divides a pie into two pieces: $(x, 1-x)$
- Player 2 observes the division and chooses one piece
- Payoffs correspond to the size of the piece each player receives

## Divide and Choose Game. Subgame Perfect Nash Equilibrium

**Step 1 (Backward Induction):**
Player 2 chooses the larger piece
If pieces are equal, Player 2 is indifferent

**Step 2 (Player 1's Problem):**
Anticipating Player 2's choice, Player 1 maximizes the size of the *smaller* piece
Optimal choice: divide the pie equally